COMPARISON GUIDE

RAG

vs

GraphRAG

How traditional retrieval-augmented generation compares to graph-based RAG for complex data and reasoning

Knowledge Graphs | Vector Search | Retrieval | LLMs | Microsoft GraphRAG

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What is RAG?

RAG (Retrieval-Augmented Generation) lets LLMs pull in information from your own data before answering. It uses vector search on text chunks so answers are grounded in real sources and hallucinations drop.

- **Ingest:** Chunk documents, turn chunks into embeddings, store in a vector DB
- **Retrieve:** Embed the user query, find the most similar chunks (e.g. cosine similarity)
- **Generate:** Send those chunks as context to the LLM and get the final answer

IN PLAIN TERMS

RAG is like giving the model a search bar over your docs. You ask a question, it finds the closest text snippets, then writes the answer using only that context.

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What is GraphRAG?

GraphRAG (from Microsoft) builds a knowledge graph from your text. It extracts entities and relationships, detects communities, and uses that structure to answer "global" questions that span the whole dataset.

- **Extract:** LLMs pull entities (people, places, orgs) and relationships from chunks
- **Graph:** Store as nodes and edges so you can reason over connections
- **Communities:** Cluster the graph (e.g. Leiden), summarize each community
- **Retrieve:** Use both graph structure and embeddings to fetch relevant subgraphs and summaries

IN PLAIN TERMS

GraphRAG turns your docs into a map of who/what is connected to what. It can answer "What themes run across the whole corpus?" not just "What does this paragraph say?"

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How They Differ

RAG

Vector search on text chunks. Best for local, fact-style questions. Fast and simple. No explicit structure.

GraphRAG

Graph over entities and relations. Best for global questions and multi-hop reasoning. Uses structure.

RAG finds "which chunks are similar to my query." GraphRAG finds "which entities and communities are relevant and how they connect."

QUICK TAKEAWAY

RAG = semantic similarity on chunks. GraphRAG = structure plus semantics for deeper, dataset-wide reasoning.

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When to Use RAG

Traditional RAG wins when the task is about finding the right passage or fact, not tracing relationships across the whole corpus.

- Simple question answering and fact lookup
- You need low latency and simpler pipelines
- Questions are well-formed and local (one doc or a few chunks)
- No strong need for "summarize everything" or "how do X and Y relate across all data"

REAL USE

Chat over docs, support bots, internal search, and most single-hop QA. Used by many production RAG systems today.

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5 When to Use GraphRAG

Repost

GraphRAG shines when you need dataset-wide understanding and relational reasoning, not just the best-matching snippet.

- Global questions: "What are the main themes across all reports?"
- Multi-hop reasoning: "How did decision A lead to outcome B via person C?"
- Discovery over private or complex data (research, legal, enterprise knowledge)
- You have (or can extract) clear entities and relationships

REAL USE

Microsoft's GraphRAG targets complex data discovery. Research shows it helps when the task needs graph structure; vanilla RAG often wins on standard QA benchmarks.

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6 **Key Insight** Repost

GraphRAG is not always better than RAG. Recent work (e.g. arXiv 2502.11371) shows they have different strengths. RAG often wins on standard QA and summarization; GraphRAG wins when the task needs relational or global reasoning.

- Choose by task and data: local facts vs global structure
- Hybrid setups (RAG + graph retrieval) are a active research direction
- Success depends on data structure, question type, and how you evaluate

BOTTOM LINE

Use RAG for "find the right paragraph." Use GraphRAG when you need "connect the dots across the whole corpus." Many teams will combine both.

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